SA-Matching DETR: A Lightweight Transformer Detector with Enhanced Scale Adaptive Matching

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Abstract

Transformer has been widely adopted in computer vision, yet Vision Transformer-based detectors often suffer from high computational costs. Furthermore, mainstream object detectors face a dilemma in label assignment: the Hungarian matching used in DETR-based models often yields false negatives, particularly for large objects, while the Intersection over Union-based matching in R-CNN models is prone to false positives with small objects. To address these challenges, we propose SA-Matching DETR, a lightweight and efficient detection framework. First, we introduce Partial ViT, a backbone that applies self-attention to partial channels via channel decomposition, significantly reducing parameters and computation while preserving rich feature representations. Second, we propose the Scale Adaptive Matching algorithm, which adaptively expands the set of ground-truth boxes based on object scale. This approach increases positive samples to mitigate false negatives without the risk of mis-assignment common in other strategies. On ImageNet-1K dataset, Partial ViT achieves 85.2% Top-1 accuracy with only 66.7M parameters, outperforming existing large-scale models. On the COCO dataset, SA-Matching DETR achieves the state-of-the-art 54.8% mAP with 84M parameters; notably, our encoder-only variant reaches 53.5% mAP, which surpasses several multi-stage detectors and demonstrates the framework's high efficiency.

1. Introduction

Transformer architecture has achieved remarkable success in Natural Language Processing (NLP) [45] and has rapidly become a cornerstone of modern computer vision. In object detection, DETR [4] marked a paradigm shift by introducing an end-to-end, anchor-free framework. Although subsequent work like Deformable DETR [57] addressed critical initial challenges such as slow convergence

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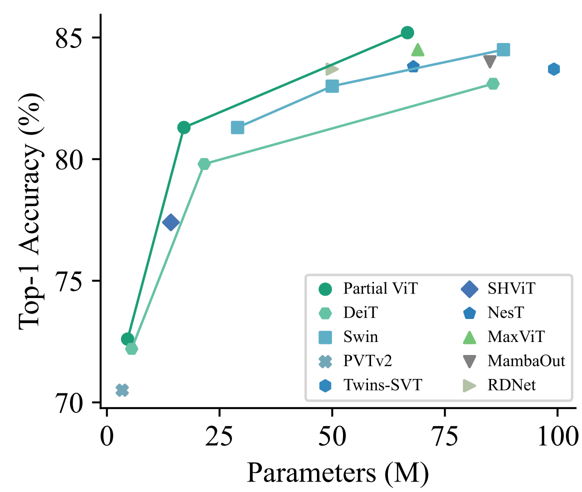


Figure 1. Comparison of Top-1 accuracy versus model size (in millions of parameters) on the ImageNet-1K dataset. Partial ViT achieves a superior accuracy-efficiency trade-off, consistently outperforming other leading Vision Transformer architectures.

and reliance on single-scale features that limited small object detection, the performance of these modern frameworks is fundamentally determined by their feature backbone. The natural successors to traditional convolutional backbones [5, 17, 20, 32, 41, 49] are Vision Transformers (ViTs) [13, 26, 31, 34, 35, 38, 42, 48], yet their integration into detection pipelines is not seamless.

The primary obstacle is the **prohibitive computational and parameter overhead** of the ViT architecture, especially in high-resolution scenarios. This overhead is a direct consequence of the global self-attention mechanism, whose cost scales quadratically with the number of input tokens. Prevailing solutions attempt to mitigate this cost through various strategies, each with inherent trade-offs. Approaches like local window attention [31] or token pruning [38] directly target the quadratic term by reducing the number of tokens in the attention operation, but risk limiting crucial long-range dependencies or discarding fine-grained

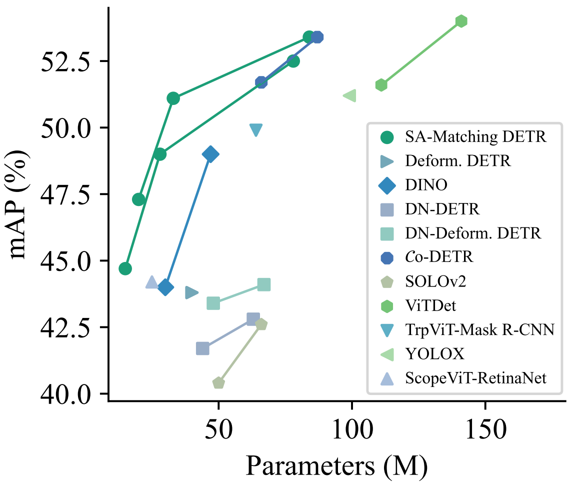


Figure 2. Comparison of mean Average Precision (mAP) against the number of parameters of various object detectors on the COCO dataset. The curve for our SA-Matching DETR consistently resides in the upper-left region of the plot, indicating a new state-of-the-art balance between detection accuracy and model complexity.

features. Other methods, such as quantization [8, 50] and low-rank decomposition [16], reduce the cost of the matrix operations within attention but may introduce feature distortion that compromises representational capacity.

To mitigate the efficiency limitations of ViT backbones, we introduce the Partial Vision Transformer (Partial ViT). This lightweight architecture employs a channel-selective attention mechanism inspired by Partial Convolution [5]. Specifically, it partitions the feature channels for each token, applying the computationally intensive self-attention operation to only one subset while the other is passed through unmodified. This strategy yields significant reductions in parameter count and computational overhead without altering the original set of tokens. By preserving the complete token sequence, Partial ViT avoids the information loss common in pruning-based methods, thus maintaining high feature integrity. This results in a superior trade-off between model efficiency and performance, which, as illustrated in Fig. 1, shows that our Partial ViT models consistently achieve higher Top-1 accuracy than competing architectures at comparable parameter counts.

Beyond backbone efficiency, a more subtle but equally critical challenge lies in the **label assignment dilemma** during training. DETR-based detectors [4, 6, 10, 30, 53, 57] typically employ Hungarian matching [24] to establish a globally optimal, one-to-one correspondence between predictions and ground-truth boxes. However, this strict constraint severely limits the number of positive samples, leading to sparse supervision and a high rate of false negatives (FNs), particularly for large objects. Conversely, traditional Intersection over Union (IoU)-based matching strategies [2, 3, 18, 33, 39, 40] are prone to generating false positives (FPs) for small objects, where a significant geometric misalignment can still produce a high IoU score. This creates a persistent trade-off that fundamentally limits detector performance.

To resolve this dilemma, we propose the Scale Adaptive Matching (SA-Matching) algorithm, a dynamic assignment strategy designed to enhance training supervision. Instead of altering the matching algorithm itself, SA-Matching adaptively expands the set of ground-truth targets based on object scale before performing the optimal one-to-one assignment. This approach enriches the gradient signals for larger objects to mitigate FNs while preserving high-precision assignments for small ones.

By synergistically integrating our lightweight backbone and advanced matching algorithm, we present SA-Matching DETR, an efficient yet powerful object detection framework. Our extensive experiments validate this approach. As illustrated in Fig. 2, the complete framework establishes a new state-of-the-art efficiency frontier on the COCO [28] object detection benchmark, consistently outperforming prior methods at comparable parameter budgets. The main contributions of this work are summarized as follows:

- We introduce **Partial ViT**, a lightweight vision transformer that reduces computational cost via channel decomposition. It achieves a superior accuracy-efficiency trade-off on image classification, attaining 85.2% Top-1 accuracy on ImageNet-1K with only 66.7M parameters.
- We propose **SA-Matching**, a label assignment strategy that resolves the false negative/positive dilemma in object detection. By adaptively creating ground-truth replicas based on object scale, it significantly enhances the density and quality of positive samples during training.
- We present the integrated **SA-Matching DETR** framework, which combines Partial ViT with SA-Matching and achieves a new state-of-the-art in efficient object detection. Our model achieves 54.8% mAP on the COCO benchmark with 84M parameters, outperforming existing models with comparable parameter budgets.

2. Related Work

Lightweight Vision Transformers. Lightweight vision transformers aim to reduce computational cost while maintaining strong feature representation. Common strategies include architectural fusion, multi-scale feature modeling, token compression, and attention simplification. Hybrid architectures merge CNNs' local feature extraction with transformers' global modeling. For example, MobileViT [34] enhances local perception using lightweight convolutions, making it well-suited for resource-limited scenarios. For multi-scale feature modeling, Swin Transformer [31] improves object-scale perception through sliding-window self-attention and a hierarchical pyramid structure. To mitigate the high computational cost of self-attention, linear attention methods have been developed, achieving near-linear complexity while maintaining competitive performance [15, 16]. Other techniques, such as Token Crop [1, 43], reduce redundant tokens based on importance scoring, effectively lowering computation. However, these approaches may discard some critical information, affecting fine-grained feature representation.

DETR-Based Object Detection Models. DETR [4] proposed a transformer-based framework for end-to-end object detection, eliminating the need for hand-designed anchors and non-maximum suppression (NMS), although it suffers from slow convergence. Deformable DETR [57] significantly improves convergence efficiency through deformable attention and is considered a landmark in the field. Building on this, numerous extensions have been proposed to enhance performance. For instance, DINO [53] employs contrastive learning to enhance discriminative power; $\mathcal{C}$o-DETR [58] uses collaborative hybrid assignment for more efficient and accurate matching; ViTDet [27] adopts a single-scale, non-hierarchical ViT backbone for efficient detection; DEYO [36] combines one-to-many detection with an end-to-end framework. To further improve detection performance, many detectors incorporate feature refinement modules [7, 54, 56], among which HPR [56] performs multi-level processing to finely enhance region features, achieving significant performance improvements.

Matching Strategies of DETR-based Detectors. Most DETR-based object detection methods perform strict one-to-one matching using the Hungarian [24] algorithm. This constraint limits the number of positive samples and readily leads to FNs for large objects. In contrast, IoU-based matching [3, 18] can generate positive samples but is highly prone to FPs. To improve matching quality, a series of methodological enhancements have been proposed. DN-DETR [25] improves detector robustness by increasing the density of positive samples through the introduction of noise queries; $\mathcal{H}$-DETR [21] integrates one-to-one and one-to-many matching schemes to enhance positive sample utilization; and Group DETR [6] accelerates convergence and strengthens detection performance by combining intra-group one-to-one with inter-group one-to-many matching. Additionally, some studies improve detection accuracy by optimizing matching quality [19, 37]. Inspired by prior advances, we propose SA-Matching, an algorithm that enhances the supervision of Hungarian matching without introducing the mis-assignment risks associated with many-to-one strategies.

3. Method

In this section, we elaborate on the proposed SA-Matching DETR, an efficient and lightweight object detection framework designed to address key limitations in existing

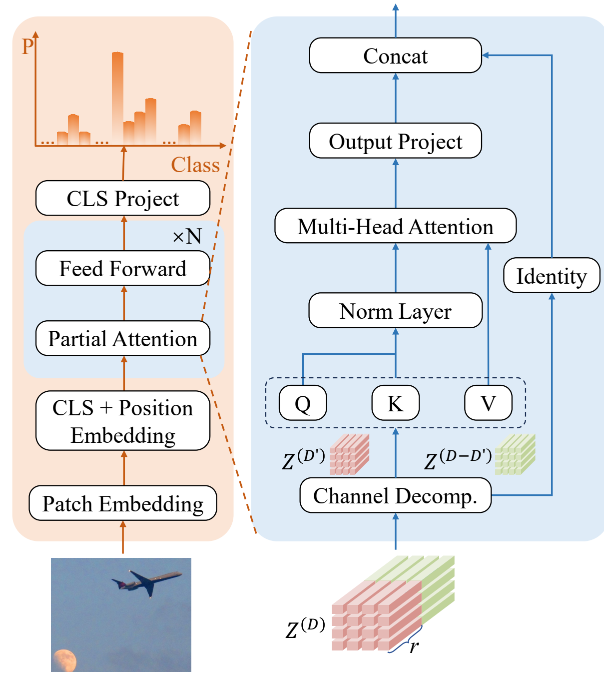


Figure 3. Architecture of the Partial ViT, highlighting the Partial Attention block (right). By decomposing features and applying multi-head attention to only a fraction of channels, the design significantly reduces computational overhead.

Transformer-based detectors. Our approach is built upon two core innovations. First, in Sec. 3.1, we introduce Partial ViT, a lightweight and efficient backbone that significantly reduces computational complexity by leveraging a channel decomposition strategy. Second, in Sec. 3.2, we present the Scale Adaptive Matching (SA-Matching) algorithm, a new label assignment strategy designed to mitigate the trade-offs between IoU-based and Hungarian matching to improve training supervision. Finally, we detail the architecture of the complete SA-Matching DETR framework in Sec. 3.3, which synergistically integrates the Partial ViT backbone to achieve a new state-of-the-art balance between performance and efficiency.

3.1. Partial Vision Transformer

Channel Decomposition. Model pruning is a common technique for creating efficient neural networks by removing redundant components. A recent example in the convolutional domain is Partial Convolution [5], a form of structured pruning that applies standard convolutions to only a subset of input channels while leaving the rest untouched. This effectively reduces both redundant computation and memory access. Inspired by this channel-wise approach, we adapt this principle to the Vision Transformer architecture. As illustrated in Fig. 3, our method, Partial ViT, introduces an efficient feature decomposition along the channel dimension. Unlike prevalent ViT compression methods that

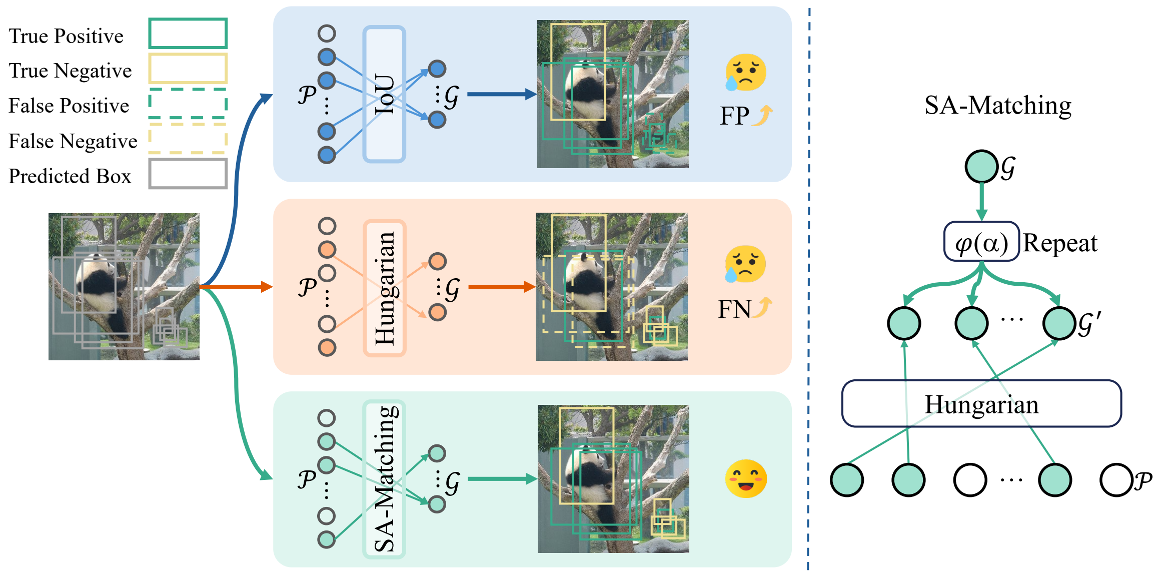


Figure 4. Comparison of matching algorithms (left) and the architecture of our proposed SA-Matching (right). SA-Matching mitigates the FPs of IoU matching and the FNs of Hungarian matching by first augmenting the ground-truth set based on object scale, then performing a one-to-one assignment.

reduce computation by pruning or merging tokens along the spatial dimension, Partial ViT retains all tokens. Instead, it applies the computationally intensive self-attention operation to only a fraction of the channels for each token. This design preserves the transformer's global semantic modeling capability and avoids the fine-grained information loss associated with token removal, achieving an effective balance between model efficiency and feature integrity.

Partial Attention. The core innovation of Partial ViT is its channel-selective partial attention block. This mechanism strategically partitions the input feature matrix along the channel dimension to reduce computational load.

Given an input feature matrix $Z \in \mathbb{R}^{N\times D}$, where N is the number of tokens and D is the feature dimension, we first define a partial channel ratio $r \in (0, 1]$. This ratio determines the number of "active" channels, D', that will undergo the self-attention operation. The input matrix Z is then split into an active partition, $Z^{(D')}$, containing the active channels, and a "frozen" partition, $Z^{(D-D')}$, containing the remaining channels, as formally defined in Eq. (1).

$$D' = \lceil rD \rceil, \quad Z = \left[Z^{(D')},\, Z^{(D-D')}\right]. \tag{1}$$

As shown in the architectural diagram (Fig. 3), these two partitions are processed through parallel pathways:

1. **Attention Pathway**: The active partition, $Z^{(D')}$, is processed by a standard transformer block. This allows the model to capture complex spatial relationships and long-range dependencies using a fraction of the original computational cost.
2. **Identity Pathway**: The frozen partition, $Z^{(D-D')}$, bypasses the expensive MHSA operation and is passed through via a residual connection. This simple yet effective step ensures that the feature information from these channels is preserved without modification.

To conclude the operation, the feature representations from both the attention pathway, $\hat{Z}^{(D')}$, and the identity pathway, $Z^{(D-D')}$, are reintegrated. This is achieved by concatenating them along the channel dimension, which produces the final output tensor $Z^{(l+1)}$ for the next block in the network:

$$Z^{(l+1)} = \text{Concat}\left(\hat{Z}^{(D')}, Z^{(D-D')}\right). \tag{2}$$

The Partial ViT design offers a compelling combination of efficiency and high performance. By applying MHSA to only a fraction of the channels, the computational cost and parameter count are substantially reduced. Our experimental results validate this design: on ImageNet-1K, the Partial ViT-B model achieves 85.2% Top-1 accuracy with only 66.7M parameters.

3.2. Scale Adaptive Matching

In modern object detectors, the label assignment strategy—the process of matching predictions to ground-truth boxes—is a critical component that governs training dynamics and ultimate model performance. The two dominant paradigms, IoU-based thresholding and Hungarian matching, each present a fundamental trade-off. IoU-based methods excel at generating a dense set of positive samples but are often imprecise, whereas Hungarian matching ensures an optimal one-to-one assignment at the cost of sparse supervision. As we will discuss, these limitations are particularly acute for objects at different scales.

IoU-Based Matching and False Positives. IoU-based assignment, widely used in R-CNN-style detectors [2, 18, 39], assigns a prediction as a positive sample if its IoU with a ground-truth box exceeds a fixed threshold. While this can generate numerous positive examples, it is highly susceptible to producing FPs, particularly for small objects. For a small target, a prediction can be significantly misaligned geometrically yet still surpass the IoU threshold, leading to the supervision of inaccurate localizations. This issue is illustrated schematically in our comparison (Fig. 4, top row).

Hungarian Matching and False Negatives. In contrast, the Hungarian algorithm [24], central to DETR-based models, establishes a globally optimal one-to-one assignment by minimizing a pairwise cost matrix. This elegant solution eliminates duplicate matches but introduces a significant side effect: FNs and sparse supervision, especially for large objects. Because of the strict one-to-one constraint, only a single prediction can be matched to each ground-truth box. For a large object, this often leaves multiple other high-quality, valid predictions unmatched, treating them as negatives during training. This mechanism is depicted conceptually in Fig. 4 (middle row), where high-quality predictions are discarded due to the one-to-one rule. Our quantitative results further confirm this, with standard Hungarian matching yielding a lower AP for large objects (58.8 $\mathrm{AP_L}$) compared to other methods (Tab. 3).

Scale Adaptive Matching. To resolve this dilemma, we propose Scale Adaptive Matching (SA-Matching), a adaptive algorithm that enhances the supervision of Hungarian matching without sacrificing its optimal assignment property or introducing the mis-assignment risks of many-to-one strategies.Let the initial set of ground-truth objects be defined as $\mathcal{G} = \{g_1, g_2, \dots, g_N\}$. We introduce a scale-aware augmentation function, $\phi(a_i)$, which determines the number of replicas to generate for each ground-truth box g_i based on its area, a_i, as formulated in Eq. (3).

$$\phi(a_i) = s + (m-s)\cdot H(a_i - \eta_s) + (l-m)\cdot H(a_i - \eta_l). \quad (3)$$

Here, $H(\cdot)$ is the Heaviside step function, which acts as a switch based on predefined area thresholds η_s and η_l. The scale-dependent augmentation factors s, m, l are hyperparameters that define the number of ground-truth copies for small, medium, and large objects, respectively. As validated by our ablation study (Fig. 5), the small-object augmentation factor is optimally set to $s = 1$. This choice maintains a one-to-one matching relationship for small targets, which is critical for mitigating the proliferation of FPs. Using the function Eq. (3), we construct an augmented ground-truth set, $\mathcal{G}'$, by duplicating each ground-truth box g_i a total of $\phi(a_i)$ times:

$$\mathcal{G}' = \bigcup_{i=1}^{N} \left\{ g_i^{(1)}, g_i^{(2)}, \dots, g_i^{(\phi(a_i))} \right\}. \quad (4)$$

During training, the standard one-to-one Hungarian matching is then performed on this expanded set $\mathcal{G}'$. This effectively enforces that medium and large objects are matched to m and l predictions, respectively. This process significantly increases the density of positive samples for larger objects, which is reflected in a substantial 6.3 AP improvement over standard Hungarian matching (Tab. 3).

3.3. The SA-Matching DETR Framework

The overall architecture of our SA-Matching DETR framework integrates the lightweight Partial ViT backbone and the advanced SA-Matching algorithm to establish a new state-of-the-art in efficient object detection. Following the design philosophy of ViTDet [27], we replace conventional hierarchical backbones with the proposed Partial ViT and derive a multi-scale feature pyramid directly from its final single-resolution output, which is then fed into a standard Transformer encoder-decoder structure. During training, label assignment between decoder predictions and ground-truth boxes is exclusively handled by our SA-Matching algorithm. The synergy between these components is critical: the Partial ViT backbone provides a highly efficient yet powerful feature extractor, the feature pyramid effectively prepares its output for multi-scale analysis, and the SA-Matching algorithm ensures dense and accurate supervisory signals, resulting in a superior balance between performance and efficiency.

4. Experiments

4.1. ImageNet-1K Classification Performance

Dataset and Evaluation Metric. Experiments on model representation capability are conducted on the ImageNet-1K dataset [11]. Following standard practice, we primarily report Top-1 classification accuracy, while also evaluating model parameter count, FLOPs, and single-GPU inference throughput to comprehensively assess model efficiency.

Implementation Details. The training pipeline is built upon the DeiT framework [42], employing the distilled DeiT models for hard distillation. Specifically, our models are initialized by trimming the weights from the pre-trained DeiT to match the reduced channel dimensions of the Partial ViT architecture; this pre-trained model then serves as the teacher network. We adopt DeiT's distillation strategy by incorporating an auxiliary distillation token that is trained to replicate the teacher's predictions. During inference, the final classification is produced by averaging the output logits from the standard class token and this distillation token. Input images are resized to 384×384 for Partial ViT-B, while the Small and Tiny variants adopt 224×224. All models are optimized using AdamW with an initial learning rate of $1\mathrm{e}{-4}$ and a minimum of $2\mathrm{e}{-6}$, following a cosine annealing schedule combined with a 5-epoch lin-

Table 1. Performance comparison of mainstream classification models on the ImageNet-1K dataset [11]. We highlight the **best** and the second-best results.

Model	Image Size	Params (M)	GFLOPs	Throughput (Img/S)	Top-1 Acc
DeiT-T [42]	224	5.5	1.1	2219.1	72.2
DeiT-S [42]	224	21.6	4.2	875.5	79.8
DeiT-B [42]	384	85.7	49.4	90.7	83.1
Twins-SVT-S [9]	224	24.0	2.8	801.5	81.7
Twins-SVT-B [9]	224	56.0	8.4	367.6	83.2
Twins-SVT-L [9]	224	99.2	14.8	246.3	83.7
Swin-T [31]	224	29.0	4.4	617.2	81.3
Swin-S [31]	224	50.0	8.5	352.7	83.0
Swin-B [31]	384	88.0	44.6	74.9	84.5
PVTv2-B0 [46]	224	3.4	0.5	1835.5	70.5
PVTv2-B2 [46]	224	25.4	3.9	558.3	82.0
PVTv2-B5 [46]	224	82.0	11.4	233.3	83.8
MaxViT-T [44]	224	31.0	5.3	355.0	83.6
MaxViT-S [44]	224	69.0	11.3	219.9	84.5
NesT-T [55]	224	17.0	5.2	449.5	81.5
NesT-S [55]	224	38.0	9.4	272.7	83.3
NesT-B [55]	224	68.0	16.7	186.2	83.8
DaViT-T [12]	224	28.3	4.4	622.7	82.8
DaViT-S [12]	224	49.7	8.6	364.4	84.2
DaViT-B [12]	224	87.9	15.2	243.2	84.6
SHViT-S1 [52]	224	6.3	0.3	1952.3	72.8
SHViT-S3 [52]	224	14.2	0.6	1760.4	77.4
SHViT-S4 [52]	256	16.5	1.0	1394.7	79.4
RDNet-T [22]	224	24.0	5.0	540.4	82.8
RDNet-S [22]	224	50.0	8.7	346.5	83.7
RDNet-B [22]	224	87.0	15.4	216.3	84.4
MambaOut-T [51]	224	27.0	4.5	595.8	82.7
MambaOut-S [51]	224	48.0	8.9	336.0	84.1
MambaOut-B [51]	224	85.0	15.8	217.8	84.2
Partial ViT-T	224	4.6	0.8	2283.5	72.6
Partial ViT-S	224	17.1	3.3	1028.0	81.3
Partial ViT-B	384	66.7	38.2	112.6	**85.2**

ear warm-up. Training is conducted for 300 epochs with a batch size of 128. Data augmentation strategies include Mixup, CutMix, Random Erasing, and multi-augmentation combinations. The classification objective is formulated as label-smoothed cross-entropy and averaged with the hard distillation loss to form the overall loss for backpropagation. Training is distributed across 8 NVIDIA A100 GPUs, while FLOPs and inference throughput are consistently measured on a single NVIDIA 2080Ti GPU.

Main Results on ImageNet-1K. Table 1 reports Partial ViT's performance across model scales and comparisons with mainstream architectures. Partial ViT-T achieves 72.6% Top-1 accuracy with 4.6M parameters, Partial ViT-S reaches 81.3%, and Partial ViT-B attains 85.2% on 384×384 inputs. Compared to DeiT-B [42], Swin [31], MaxViT [44], and MambaOut [51], Partial ViT offers higher efficiency while maintaining accuracy. Specifically, against DeiT-B, Top-1 improves by 2.1% with 22% fewer parameters, 23% lower FLOPs, and 24% higher throughput; versus Swin-B, accuracy rises 0.7% with 24% and 14% reductions in parameters and FLOPs, and 50% higher throughput. These results demonstrate that Partial ViT effectively balances compact design and high precision.

4.2. COCO Object Detection Performance

Dataset and Evaluation Metric. Experiments on object detection performance are conducted on the COCO benchmark [28]. Following standard practice, we report AP on the validation set, including AP_{50}, AP_{75}, AP_{S}, AP_{M}, AP_{L}, for comprehensive evaluation.

Implementation Details on COCO. The training pipeline is built upon the MMDetection [57] framework. The detector is initialized with weights pre-trained on ImageNet-1K [11] and adopts a DETR-based architecture with 900 object queries for prediction generation. Optimization is performed using the AdamW optimizer [42], with a total batch size of 16. Input images are resized to 1024×1024, and the learning rate is set to $2\mathrm{e}{-4}$ with a linear warm-up of 250 iterations. For the 24-epoch training schedule, the learning rate decays by a factor of 0.1 at epoch 20 (and at epoch 11 for the 12-epoch schedule). Standard DETR-style data augmentations [21, 53] are employed, combined with

Table 2. Performance comparison of mainstream object detection models on the COCO dataset [28]. All models were trained on 8 NVIDIA A100 (40GB) GPUs. We highlight the **best** and the second-best results.

Model	Backbone	Epochs	mAP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params (M)
One-stage Detectors									
FoveaBox [23]	ResNet-101	12	40.8	61.4	44.0	24.1	45.3	53.2	55
FoveaBox [23]	ResNeXt-101	12	42.3	62.9	45.4	25.3	46.8	55.0	58
RetinaNet [29]	ScopeVIT-T [35]	12	41.9	63.0	44.7	27.4	45.8	56.5	17
RetinaNet [29]	ScopeVIT-S [35]	12	44.2	65.5	47.3	27.5	48.3	59.2	25
SOLOv2 [47]	ResNet-50	72	40.4	59.8	42.8	20.5	44.2	53.9	50
SOLOv2 [47]	ResNet-101	72	42.6	61.2	45.6	22.3	46.7	56.3	66
SOLOv2 [47]	ResNet-101-DCN	72	44.9	63.8	48.2	23.1	48.9	61.2	69
YOLOX-M [14]	ModifiedCSPv5	300	46.4	65.4	50.6	26.3	51.0	59.9	25
YOLOX-L [14]	ModifiedCSPv5	300	50.0	68.5	54.5	29.8	54.5	64.4	54
YOLOX-X [14]	ModifiedCSPv5	300	51.2	69.6	55.7	31.2	56.1	66.1	99
SA-Matching DETR Enc.	Partial ViT-T	12	44.7	61.8	49.1	25.4	48.6	61.7	15
SA-Matching DETR Enc.	Partial ViT-S	12	49.0	67.2	53.5	29.6	53.2	67.5	28
SA-Matching DETR Enc.	Partial ViT-B	12	52.5	71.7	57.2	32.2	57.3	70.8	78
SA-Matching DETR Enc.	Partial ViT-T	24	47.6	65.2	56.1	27.7	51.6	65.4	15
SA-Matching DETR Enc.	Partial ViT-S	24	51.3	69.7	52.1	32.0	55.9	69.1	28
SA-Matching DETR Enc.	Partial ViT-B	24	**53.5**	**73.0**	**59.0**	**34.8**	**58.7**	**72.4**	78
Multi-stage Detectors									
DETR [4]	ResNet-50	500	42.0	62.4	44.2	20.5	45.8	61.1	41
DETR [4]	ResNet-101	500	43.5	63.8	46.4	21.9	48.0	61.8	60
DAB-DETR [30]	ResNet-50	50	42.2	63.1	44.7	21.5	45.7	60.3	44
DAB-DETR [30]	ResNet-101	50	43.5	63.9	46.6	23.6	47.3	61.5	63
Sparse R-CNN [40]	ResNet-50	36	42.8	61.2	45.7	26.7	44.6	57.6	106
Sparse R-CNN [40]	ResNet-101	36	44.1	62.1	47.2	26.1	46.3	59.7	125
Deform. DETR [57]	ResNet-50	50	46.2	65.2	50.0	28.8	49.2	61.7	40
DN-DETR-DC5 [25]	ResNet-50	12	41.7	61.4	44.1	21.2	45.0	60.2	44
DN-DETR-DC5 [25]	ResNet-101	12	43.4	61.9	47.2	24.8	46.8	59.4	63
DN-Deform. DETR [25]	ResNet-50	12	43.4	61.9	47.2	24.8	46.8	59.4	48
DN-Deform. DETR [25]	ResNet-101	12	44.1	62.8	47.9	26.0	47.8	61.3	67
Mask R-CNN [18]	TrpViT-T [26]	36	48.7	71.0	53.5	–	–	–	47
Mask R-CNN [18]	TrpViT-S [26]	36	49.9	72.1	55.2	–	–	–	64
Mask R-CNN [18]	TrpViT-B [26]	36	50.5	72.4	55.1	–	–	–	99
DINO-4scale [53]	ResNet-50	12	49.0	66.6	53.5	32.0	52.3	63.0	47
DINO-4scale [53]	ResNet-50	24	50.4	68.3	54.8	33.3	53.7	64.8	47
$\mathcal{C}$o-DETR [58]	ResNet-50	12	49.5	67.6	54.3	32.4	52.7	63.7	65
$\mathcal{C}$o-DETR [58]	Swin-T	12	51.7	69.6	56.4	34.4	54.9	66.8	66
$\mathcal{C}$o-DETR [58]	Swin-S	12	53.4	71.8	58.5	35.4	57.0	69.0	87
ViTDet Mask [27]	ViT-B	100	51.6	72.1	56.6	35.3	55.6	66.4	111
ViTDet Cascade [27]	ViT-B	100	54.0	–	–	–	–	–	141
SA-Matching DETR	Partial ViT-T	12	47.3	64.9	51.2	28.3	50.5	64.8	20
SA-Matching DETR	Partial ViT-S	12	51.1	69.5	55.7	31.4	55.0	68.8	33
SA-Matching DETR	Partial ViT-B	12	53.4	72.5	58.1	34.3	58.0	71.4	84
SA-Matching DETR	Partial ViT-T	24	50.7	69.0	55.1	31.6	54.0	68.0	20
SA-Matching DETR	Partial ViT-S	24	53.3	72.0	58.2	34.6	57.4	70.6	33
SA-Matching DETR	Partial ViT-B	24	**54.8**	**73.8**	**59.5**	**36.4**	**59.3**	**72.3**	84

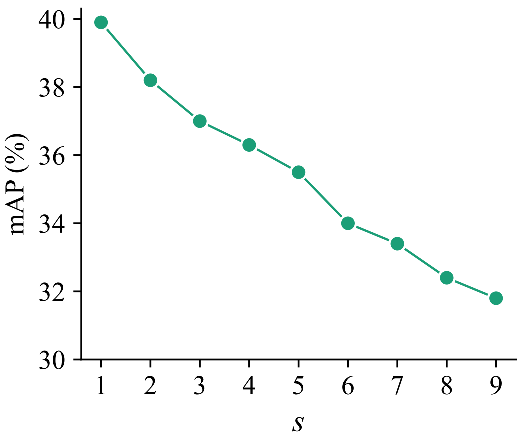
Re-aug [56], to enhance training robustness.

Main Results on COCO. Table 2 presents the performance of SA-Matching DETR across model scales and comparisons with mainstream architectures. The SA-Matching DETR framework demonstrates state-of-the-art performance, establishing a new standard for accuracy and efficiency. Our lightweight, encoder-only models are highly competitive; for instance, the 28M parameter Partial ViT-S version achieves 51.3% mAP, surpassing the much larger YOLOX-X [14]. By integrating 3 decoders, our full framework sets a new state-of-the-art, with the 84M parameter Partial ViT-B model reaching 54.8% mAP. This result exceeds the performance of strong contemporary models like $\mathcal{C}$o-DETR [58]. These findings validate that our framework excels at high-performance, lightweight object detection.

4.3. Ablation Studies

In this section, we conduct a series of ablation studies to validate our key design choices and quantify the contribution of each component within the SA-Matching DETR framework. Our experiments are conducted using the **Partial ViT-S** backbone and 12 epochs unless otherwise stated.

Scale-dependent Augmentation Factors. We conducted an ablation study to determine the optimal values for the scale-dependent augmentation factors (s, m, l). These early-stage experiments were conducted with a ResNet-50 backbone to establish a robust set of hyperparameters before proceeding with our proposed model. As illustrated in Fig. 5, the mAP is inversely correlated with the small-object factor, s. Performance is maximized at $s = 1$, which aligns



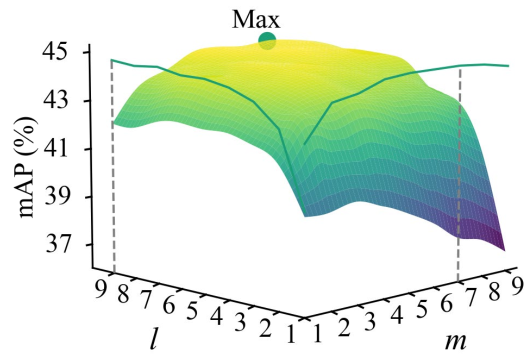


Figure 5. Ablation study on the small-object augmentation factor, s. Performance peaks at $s = 1$, indicating that a one-to-one assignment is optimal for small objects.

Figure 6. Performance landscape for the medium (m) and large (l) object augmentation factors. The empirical results show a distinct optimal region, with performance peaking at $(m, l) = (7, 9)$.

Table 3. Detection performance under SA-Matching, Hungarian, and IoU matching algorithms.

Algorithm	$\mathrm{AP_S}$	$\mathrm{AP_M}$	$\mathrm{AP_L}$	mAP
IoU	23.8	50.7	63.7	44.6
Hungarian	25.0	46.0	58.8	42.7
SA-Matching	29.6	53.2	67.5	49.0

Table 4. Ablation study of framework components on ResNet-50 and Partial ViT-S backbones.

Backbone	Partial Attn.	Data Re-aug	SA-Matching	mAP
ResNet-50	–			39.8
	–	✓		41.2
	–		✓	44.7
	–	✓	✓	46.0
ViT-S				32.8
	✓			34.4
	✓	✓		42.7
	✓		✓	42.8
	✓	✓	✓	49.0

with our goal of mitigating FPs. For medium and large objects, the performance landscape is visualized in Fig. 6. The empirical results show a clear optimal region, with the peak mAP achieved when the augmentation factors are set to $m = 7$ and $l = 9$. Based on this analysis, we adopt the configuration of $(s, m, l) = (1, 7, 9)$ as the default setting for our SA-Matching algorithm.

SA-Matching Strategy Analysis. The SA-Matching algorithm is designed to integrate the strengths of both Hungarian and IoU-based matching to improve the quality of positive sample assignment during training. As shown in Tab. 3, our SA-Matching algorithm significantly outperforms both conventional strategies on the COCO benchmark. It achieves a remarkable 6.3 point mAP increase over the standard Hungarian algorithm and a 4.4 point mAP increase over IoU-based matching. This superior performance is consistent across all object scales, with notable gains in $\mathrm{AP_S}$, $\mathrm{AP_M}$, and $\mathrm{AP_L}$ demonstrating the versatility of our approach.

Component Analysis. To comprehensively assess each module's contribution, we conducted an ablation study on both ResNet-50 and our proposed ViT-S backbones (Tab. 4). As shown in the table, the Data Re-aug strategy provides a modest 1.3% mAP gain (44.7% $\rightarrow$ 46.0%) when applied to a ResNet-50 backbone with SA-Matching. In contrast, its impact is far more dramatic on our architecture, where it delivers a substantial 6.2% mAP improvement (42.8% $\rightarrow$ 49.0%) for the Partial ViT-S backbone combined with SA-Matching. This pronounced difference demonstrates a strong synergistic effect between our proposed backbone, our matching algorithm, and the augmentation technique.

5. Conclusion

In this work, we introduce SA-Matching DETR, which synergistically integrates the lightweight Partial ViT backbone with the advanced SA-Matching algorithm to establish efficient object detection. Partial ViT applies multi-head self-attention to only a fraction of channels, substantially reducing computational cost and parameter count. Meanwhile, SA-Matching strengthens Hungarian matching supervision while preserving its optimal assignment property and avoiding the pitfalls of many-to-one strategies. Experiments show that Partial ViT achieves 85.2% Top-1 accuracy with only 66.7M parameters on ImageNet, while SA-Matching DETR attains a state-of-the-art 54.8% mAP on COCO with 84M parameters, demonstrating remarkable efficiency.

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